

Sales & Profitability Analysis

Global Electronics Retailer — End-to-End Business Analysis Project

Excel • MySQL • Power BI

\$57.5M Total Revenue Analyzed	\$32.7M Total Profit	62,884 Transactions	5 Years 2016 – 2021
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Executive Summary

This project analyzes five years of transaction data from a global electronics retailer to identify which products, stores, and time periods drive the most profit. Using a raw, uncleaned Kaggle dataset, I performed the full analytics lifecycle myself: data cleaning in Excel, relational database design and querying in MySQL, and an interactive Power BI dashboard — arriving at findings that would directly inform inventory, marketing, and channel investment decisions for a business like this.

Business Objective

Analyze historical sales performance to identify which products, stores, and time periods drive the most profit, in order to guide future inventory and store decisions. The analysis was scoped to support two audiences: a Sales Director needing a company-wide view, and Store Managers needing to see their own location's performance specifically.

Tools & Skills Demonstrated

Excel: data auditing, cleaning text/number/date inconsistencies, formula-driven transformation	SQL (MySQL): relational schema design, composite keys, multi-table joins, aggregation
Power BI: data modeling, DAX measures, interactive dashboards with slicers	Analysis: KPI definition, currency conversion logic, time-series trend analysis

Data Overview

The dataset consists of five related tables sourced from Kaggle's "Global Electronics Retailer" dataset:

- Sales — 62,884 individual line items (Jan 2016 – Feb 2021)
- Products — 2,517 SKUs with cost and price
- Customers — 15,266 customers across multiple countries
- Stores — 68 locations (67 physical + 1 online channel)
- Exchange_Rates — daily currency conversion rates for multi-currency sales

Methodology

- **1. Data Cleaning (Excel):** audited all five source files for structural and formatting issues — inconsistent date formats, currency-formatted text fields, and encoding inconsistencies — before any analysis began.

- **2. Database Design & Analysis (MySQL):** designed a normalized star-schema database (one fact table, four dimension tables) with proper primary/foreign key relationships, then wrote SQL queries to answer each core business question, validating each result against real-world expectations before treating it as final.
- **3. Dashboard Development (Power BI):** built a relational data model, developed reusable DAX measures matching the validated SQL logic, and designed a three-page interactive report tailored to two distinct audiences.

Key Findings

1. Profitability is concentrated in high-value electronics

Desktop PCs and large-screen TVs account for the highest total profit by a wide margin, while accessories (USB cables, case fans) contribute the least per product. No products were sold at a loss.

2. Desktops lead in both profit and volume

The same desktop PC product lines that top the profit ranking also lead in units sold (500+ units each) — indicating this category is the business's core strength on every measure, not just margin.

3. Online sales rival entire national markets

While the United States leads overall (driven by its large number of physical stores), the Online channel alone generates more revenue than entire countries such as the United Kingdom or Germany — highlighting e-commerce as a major, underleveraged growth channel rather than a secondary sales path.

4. Clear, recurring seasonality

Revenue spikes every December without exception across all five years, consistent with holiday shopping demand — a reliable, actionable pattern for inventory and staffing planning.

5. A sharp, attributable revenue disruption in 2020

Revenue fell approximately 90% between February and April 2020, coinciding precisely with the onset of COVID-19 lockdowns, with only a partial recovery by early 2021. This is the most significant single pattern in the dataset and demonstrates how external shocks are directly visible in transaction-level data.

6. Revenue skews heavily toward older customers

Customers aged 46 and older account for roughly 60-70% of total revenue, far outweighing every younger age bracket combined. This is a strong signal for retention-focused marketing over youth-targeted campaigns, and suggests loyalty and repeat-purchase programs may offer more return than acquisition spend aimed at younger demographics.

7. Customer geography closely tracks store geography, with one notable gap

Revenue by customer country largely mirrors revenue by store country (United States, United Kingdom, and Germany lead both views), confirming the store network is generally well-aligned with where demand exists. The main divergence comes from customers in a store's country purchasing through the Online channel rather than a physical location — a pattern worth further investigation for channel strategy.

Data Quality & Problem-Solving

The source data required substantial cleaning before analysis was reliable: inconsistent character encodings across files, currency values stored as formatted text, non-standard date formats, and missing delivery records (left intact and treated as a legitimate data feature rather than an error, since they represented ~79% of orders — likely reflecting untracked or in-store pickup transactions rather than data loss). Multi-currency revenue figures were converted to a common USD basis using date-specific exchange rates. Each calculation was independently validated by cross-checking SQL and Power BI outputs against each other before being treated as reliable.

Dashboard Overview

- **Overview page:** company-wide KPI cards, a monthly revenue trend line (surfacing the seasonality and COVID-19 findings), and a revenue-by-country breakdown.
- **Store Performance page:** interactive Country/State slicers letting any store manager isolate their own location's performance in real time.
- **Product Performance page:** top-10 rankings by profit and by quantity sold, side by side.
- **Customer Performance page:** revenue by customer country and by age group, surfacing the retention-focused targeting opportunity.

Business Recommendations

- Prioritize inventory and marketing investment toward desktop PCs and large-screen TVs, the clear leaders on both profit and volume.
- Treat the Online channel as a strategic growth priority, given its revenue already rivals entire physical markets.
- Plan inventory and staffing ahead of the consistent December demand spike.
- Investigate the low-performing lighting/accessory category to determine whether it merits continued shelf space.
- Shift marketing spend toward retention and loyalty programs for the 46+ customer base, which drives the large majority of revenue, rather than defaulting to youth-oriented acquisition campaigns.

Conclusion

This project demonstrates the ability to take a real, imperfect dataset from raw files through to a decision-ready interactive dashboard — independently handling data cleaning, relational database design, SQL analysis, and BI development, while validating findings at every stage rather than presenting numbers at face value.